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Software Engineer

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SUMMARY

2026 Information Technology graduate (PICT, CGPA 8.94/10) building secure, scalable distributed systems in Java and Python. Solo-architected FinTwin.ai end to end — four services, Kubernetes deployment, ~180 automated tests. Strong fundamentals in DSA, OS, DBMS and networking, with two published research papers.

WORK EXPERIENCE

Google AI/ML Virtual Internship

Eduskills, Virtual | 01/2025 – 04/2025

- Implemented ML pipelines for classification and NLP using **CNN and XGBoost** across 50,000+ records, improving accuracy 15–20%.
- Designed **RESTful APIs** for secure inter-service transmission, improving transfer efficiency by 30%.

Yoga Pose Detection Internship

Pune Institute of Computer Technology, Pune | 02/2025 – 04/2025

- Developed a real-time posture analysis system using **MediaPipe, CNN, XGBoost and SVM** — 20% accuracy gain, 35% better error detection, 25% faster processing.
- **Led a 3-member team** across the full lifecycle — design, deployment, testing, performance tuning and UX.
- Integrated secure communication and **fault-tolerant** response design for live-streaming analytics.

PROJECTS

FinTwin.ai — AI-Powered Personal Finance Platform

Self-driven, 2026 | github.com/SYEDMDSAAD/FinTwin.ai

Solo-built platform: bank-account aggregation via **Setu AA** with budget, goal, investment and net-worth engines featuring real **EMI & DTI** computation and a water-fill savings-allocation algorithm.

- **Architected** a production-grade 4-service distributed system — **React + Vite** frontend, **Spring Boot (Java 21)** core API, a separate identity service and a **Python/FastAPI** AI service over **PostgreSQL** with 14 **Flyway** migrations.
- **Engineered JWT** auth with refresh-token rotation and theft detection, TOTP 2FA, **Google OAuth**, RBAC via a custom Spring Security permission evaluator, and **AES-256-GCM** field-level encryption of financial data.
- **Deployed** a self-hosted **LLM** copilot on **Ollama** (qwen2.5 / phi3) using an OpenAI-style tool-calling loop that fetches user data through a secured internal API rather than prompt-stuffing — no paid API dependency.
- **Eliminated** N+1 query hotspots through **Redis**-backed distributed rate limiting, Caffeine caching, HikariCP tuning, paginated batch jobs replacing full-table scans, and parallel market-data fetching.
- **Shipped** on **Docker** and **Kubernetes** (HPA 2→10 replicas, zero-downtime rolling deploys), **Prometheus/Grafana** monitoring, 5 **CI/CD** workflows and ~180 tests (48 **Testcontainers**, 129 pytest).

Real-Time Chat App

Built a **Socket.io** chat app with real-time one-to-one and group messaging and delivery-status tracking.

- Secured sessions with **JWT** and bcrypt; achieved <50 ms message delivery across concurrent sessions.

Sentiment Analysis Service

Spring Boot microservice on **AWS Comprehend** for real-time sentiment detection via asynchronous, non-blocking **REST APIs**.

EDUCATION

Information Technology
Pune Institute of Computer Technology
11/2022 – 06/2026
CGPA: 8.94 / 10.0

HARD SKILLS

- Java, Python, SQL, Spring Boot, FastAPI, Node.js, Express.js, React.js
- PostgreSQL, MySQL, MongoDB, Redis, Flyway
- Docker, Kubernetes, AWS (Lambda, EC2, S3, API Gateway), GCP, CI/CD, Prometheus/Grafana, Git
- DSA, OOP, Distributed Systems, Microservices, OS, DBMS, Networking, REST APIs, JWT/OAuth2, AES-256 Encryption, RBAC, Performance Optimization, Testcontainers, LLM/Ollama

ACHIEVEMENTS

IEEE Publication

"*Yoga Pose Detection and Correction Using Machine Learning Approaches*", ICCCNT, IIT Indore, 2025. Real-time posture correction model using CNN, XGBoost and MediaPipe, achieving 92% pose accuracy.

Journal Publication

"*Machine Learning Techniques for Detecting DoS and DDoS Attack: Methods, Attacks and Datasets*", Indian Journal of Technical Education (ISTE), 2026. Intrusion detection system using MLP and LSTM, achieving >99% accuracy on CIC-IDS benchmarks.